Sensitivity-Gradient Tipping, MPC-Guided Policy Search and Suspension Integrity for Cable-Only Locomotion of a Six-Bar Tensegrity Rover

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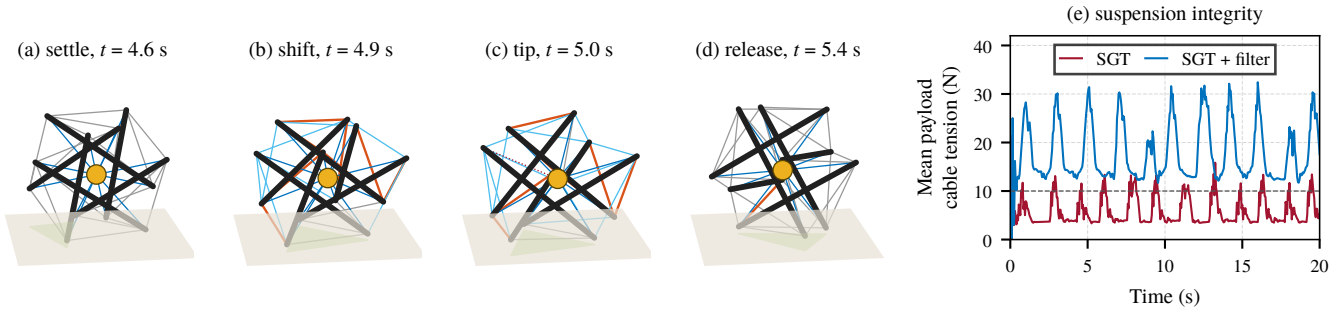


Fig. 1. One cable-only tip of the six-bar rover under Sensitivity-Gradient Tipping with the suspension-integrity filter, rendered from the simulator: (a) settled on a support face (green), (b) the cables selected by the sensitivity contract (orange) or lengthen (light blue) and shift the payload (yellow), (c) the centre of mass crosses the support edge, (d) cables released on the new face. Payload cables are blue when taut and dotted red when slack. (e) Mean payload-cable tension over 20 s: without the filter the suspension stays below 10 N (dashed) and is partly slack; with it the payload remains suspended.

***Abstract*—Six-bar tensegrity rovers roll by deforming their cable network, but their gaits are developed in simulators that can inject energy, and their suspended payloads are rarely examined. We present three cable-only controllers on an energy-audited simulator. Sensitivity-Gradient Tipping (SGT) differentiates the pinned quasi-static equilibrium and reaches 0.091 m/s, where support-face LQR, iLQR and MPC schedulers advance at most 0.10 m in 20 s. MPC + GPS, an MPC-guided policy search whose teacher reduces exactly to SGT, learns a cable policy reaching 89 % of SGT's progress on held-out conditions. A suspension-integrity filter with discrete-time barrier conditions reduces payload-cable slack from 35.1 % to 2.7 % of the time and increases SGT's progress by 48 %. Code: https://github.com/kapildev1012/3d.**

I. Introduction

Six-bar spherical tensegrities combine low mass with high impact tolerance and have been studied as planetary landers and rovers [1], [2]. They move by *punctuated rolling*: changing cable rest lengths deforms the shell until the centre of mass (COM) leaves the triangular support face and the structure tips onto a neighbouring face (Fig. 1). Because hardware is expensive and fragile, gaits are developed in simulation [3], [4] and transferred later, which makes two questions central: does the simulated motion come from the cables, and does the gait leave the payload it carries intact?

General engines such as MuJoCo and Bullet [5], [6] and dedicated tensegrity simulators [3] treat bars as rigid bodies or stiff penalty springs whose drift is removed by stabilization [7] or by position projection (SHAKE/RATTLE [8], [9]); contacts are often clamped, and interactive simulators add "assist" torques so that a gait looks convincing. Each of these changes the mechanical energy without a physical work term, and structure-preserving integrators [10] do not by themselves expose such sources. On the control side, rolling gaits have been obtained from hand-designed cable sequences, kinodynamic planning over gait primitives [2] and reinforcement learning, including guided policy search on SUPERball [4]. Guided policy search trains a policy by supervised learning on local trajectory optimizers that are kept close to it [11], [12]; its MPC-guided variant uses a model-predictive teacher [13], and PLATO and DAgger train on the learner's own state distribution [14], [15]. Control barrier functions enforce safety by a minimal correction of a nominal input [16], [17], but we are not aware of their use for the suspension of a tensegrity payload. Classic support-face gaits, finally, contract a fixed cable set; with a realistic stroke and a heavy payload such a primitive may not move the COM far enough to tip, however its effort is scheduled.

Building on an *energy-audited* simulator—a non-minimal Lagrangian model [18] with exact bar multipliers, passive contact and a work ledger that measures, before correction, the energy that each step creates without a booked cause—we make three contributions (Fig. 2): (1) *Sensitivity-Gradient Tipping* (SGT), a cable-only gait that differentiates the pinned quasi-static equilibrium of shell and payload, including the geometric stiffness of the compressed bars, and drives the COM across a support edge; (2) *MPC + GPS*, an MPC-guided policy

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search—throughout, GPS denotes Guided Policy Search—whose receding-horizon QP teacher on the SGT sensitivity reduces exactly to SGT in a limit and trains a permutation-equivariant per-cable policy; and (3) the observation that the payload suspension of a rolling six-bar rover loses its pretension, with a *suspension-integrity filter* that minimally corrects any tipping controller using the same sensitivities. All methods are evaluated on a seed-locked protocol (114 runs for SGT and five baselines, 24 for policy training and evaluation, 40 for the filter), and every number in the text is generated from the run files. All claims are restricted to simulation, and the negative results for the baselines concern the support-face primitive they schedule, not the optimal-control methods as such.

II. Energy-Audited Shell Dynamics

A. Coordinates, Mass and Constraints

The shell has $n = 12$ nodes $\boldsymbol{q} \in \mathbb{R}^{36}$, six bars connecting node pairs (i, j) and 24 outer cables; the payload core is a point mass m_c suspended by 12 cables. Bar k of length l_k, mass m_k and radius r_k is a uniform cylinder, which gives the consistent block mass matrix (times I_3)

$$\boldsymbol{M}_k = \begin{bmatrix} a_k + m_i & b_k \\ b_k & a_k + m_j \end{bmatrix},\ a_k = \tfrac{m_k}{4} + \tfrac{J_k}{l_k^2},\ b_k = \tfrac{m_k}{4} - \tfrac{J_k}{l_k^2}, \tag{1}$$

with $J_k = \frac{m_k}{12}(3r_k^2 + l_k^2)$ and node masses m_i. The bars are node-disjoint, so $\boldsymbol{M}$ is block diagonal with closed-form inverse. The bar constraints are $R_k(\boldsymbol{q}) = \|\boldsymbol{b}_k\|^2 - l_k^2 = 0$, $\boldsymbol{b}_k = \boldsymbol{q}_j - \boldsymbol{q}_i$, with Jacobian $\boldsymbol{R}_q$.

B. Equations of Motion and Correction

With $\boldsymbol{F}$ collecting gravity, cable, damping, contact and payload forces, the dynamics are $\boldsymbol{M}\ddot{\boldsymbol{q}} = \boldsymbol{F} + \boldsymbol{R}_q^\top \boldsymbol{\lambda}$ with $\boldsymbol{R}(\boldsymbol{q}) = \boldsymbol{0}$. Differentiating the constraints twice, $\ddot{R}_k = 2(\dot{\boldsymbol{b}}_k^\top \dot{\boldsymbol{b}}_k + \boldsymbol{b}_k^\top \ddot{\boldsymbol{b}}_k) = 0$, i.e. $\boldsymbol{R}_q \ddot{\boldsymbol{q}} + \boldsymbol{\xi}_2 = \boldsymbol{0}$, and substituting $\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}(\boldsymbol{F} + \boldsymbol{R}_q^\top \boldsymbol{\lambda})$ gives the index-1 multipliers in closed form,

$$\boldsymbol{\lambda} = -\big(\boldsymbol{R}_q \boldsymbol{M}^{-1} \boldsymbol{R}_q^\top\big)^{-1}\big(\boldsymbol{\xi}_2 + \boldsymbol{R}_q \boldsymbol{M}^{-1} \boldsymbol{F}\big), \quad \xi_{2,k} = 2\|\dot{\boldsymbol{b}}_k\|^2, \tag{2}$$

and the bar compression $P_k = 2\lambda_k l_k$ ($\Delta t = 2\,\mathrm{ms}$, semi-implicit Euler). After every step the minimum-norm corrector of [18] iterates, for at most ten iterations,

$$\boldsymbol{r} = \big[\boldsymbol{R};\ \dot{\boldsymbol{R}};\ E + V_c - (E^\star + \delta W)\big], \tag{3a}$$

$$\boldsymbol{d} = -\boldsymbol{A}^\top(\boldsymbol{A}\boldsymbol{A}^\top)^{-1}\boldsymbol{r}, \quad \boldsymbol{A} = \partial \boldsymbol{r}/\partial[\boldsymbol{q}; \dot{\boldsymbol{q}}; \delta W], \tag{3b}$$

$$[\boldsymbol{q}; \dot{\boldsymbol{q}}; \delta W] \leftarrow [\boldsymbol{q}; \dot{\boldsymbol{q}}; \delta W] + \boldsymbol{d}, \tag{3c}$$

where $E^\star$ is the energy predicted by the work ledger and the correction work δW is booked explicitly instead of being hidden. Our extension is the contact potential V_c in the energy function. The energy row is enforced only while $|E - E^\star| \le \max(0.5\,\mathrm{J}, 0.02|E|)$, so a gross modelling error is reported rather than corrected.

C. Passive Contact

For penetration $p = \max(0, h(x, y) - z)$ into terrain h, the normal force is the Hunt–Crossley law [19] $\boldsymbol{f}_n = k\, p^{3/2}(1 + \alpha \dot{p})\hat{\boldsymbol{n}}$, whose elastic part has potential $V_c = \frac{2}{5} k\, p^{5/2}$ and whose damping only dissipates; friction is a regularized Coulomb law $\boldsymbol{f}_t = -\mu \|\boldsymbol{f}_n\| \boldsymbol{v}_t / \sqrt{\|\boldsymbol{v}_t\|^2 + \epsilon^2}$. No clamps, impulses or velocity projections are used.

D. Work Ledger

The ledger stores the energy

$$E(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \tfrac{1}{2}\dot{\boldsymbol{q}}^\top \boldsymbol{M} \dot{\boldsymbol{q}} - \textstyle\sum_i m_i \boldsymbol{g}^\top \boldsymbol{q}_i + \sum_c \tfrac{2}{5} k\, p_c^{5/2} \tag{4}$$

and, for every force category c, the trapezoidal work $W_c \mathrel{+}= \frac{\Delta t}{2} \boldsymbol{F}_c^\top (\dot{\boldsymbol{q}}_k + \dot{\boldsymbol{q}}_{k+1})$; projections are booked as the energy change they cause. The certificate and the per-step drift are

$$\varepsilon(t) = E(t) - E_0 - \textstyle\sum_c W_c(t), \qquad \delta_k = \varepsilon_k^- - \varepsilon_{k-1}, \tag{5}$$

where ε_k^- is the residual before the energy row is applied: δ_k is the energy that step k created or destroyed without a booked cause.

Proposition 1 (First-order drift): Let all forces be booked and Lipschitz along the trajectory, and let the semi-implicit Euler step have local error $O(\Delta t^2)$. Then $\delta_k = O(\Delta t^2)$ and $\sum_{k \le T/\Delta t} |\delta_k| = O(\Delta t)$. An unbooked force $\boldsymbol{f}_u$ instead contributes $\sum_k |\delta_k| \to \int_0^T |\boldsymbol{f}_u^\top \dot{\boldsymbol{q}}|\, dt \neq 0$ as $\Delta t \to 0$.

Proof: Along the exact flow $\dot{E} = \sum_c \boldsymbol{F}_c^\top \dot{\boldsymbol{q}}$, so the exact increment equals the exact work. The trapezoidal work is exact up to $O(\Delta t^3)$ for smooth integrands, and E evaluated at the numerical state differs from its exact value by the local error times ∇E, i.e. $O(\Delta t^2)$. Summing $T/\Delta t$ steps gives $O(\Delta t)$. An unbooked power $\boldsymbol{f}_u^\top \dot{\boldsymbol{q}}$ is a Riemann sum that converges to its integral instead of vanishing. ∎

Halving Δt must therefore halve $\sum_k |\delta_k|$ for a correct model; Section VI verifies this. We also report the numerical work $|W_{\mathrm{num}}|$ of projections and corrector and the *aid share* of the energy input supplied by non-physical categories. A run is *admissible* if the aid share is zero and $\max_k |\delta_k|$ and $|W_{\mathrm{num}}|$ are small compared with the actuator work.

III. Sensitivity-Gradient Tipping

A. Quasi-Static Sensitivity

Between tips the rover rests on a support face $\mathcal{F}$ of three sticking nodes. Augmenting $\boldsymbol{q}$ with the core ($n' = 13$ nodes) and linearizing the static equilibrium with respect to the rest-length offsets $\boldsymbol{u} \in \mathbb{R}^{36}$ ($u > 0$ shortens) gives

$$\begin{bmatrix} \boldsymbol{K} & \boldsymbol{A}^\top \\ \boldsymbol{A} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \delta \boldsymbol{q} \\ \delta \boldsymbol{\mu} \end{bmatrix} = \begin{bmatrix} \boldsymbol{B} \\ \boldsymbol{0} \end{bmatrix} \delta \boldsymbol{u}, \quad \boldsymbol{A} = \begin{bmatrix} \boldsymbol{R}_q \\ \boldsymbol{E}_{\mathcal{F}} \end{bmatrix}, \tag{6}$$

where $\boldsymbol{E}_{\mathcal{F}}$ pins the support nodes and $\boldsymbol{B}$ contains $\pm k\boldsymbol{e}$ for every taut cable. $\boldsymbol{K}$ assembles $k\boldsymbol{e}\boldsymbol{e}^\top + \frac{t}{L}(\boldsymbol{I} - \boldsymbol{e}\boldsymbol{e}^\top)$ for a taut cable and the *negative* geometric stiffness $-\frac{P}{L}(\boldsymbol{I} - \boldsymbol{e}\boldsymbol{e}^\top)$ for a bar under compression P; axial bar rigidity is a constraint, and a small Tikhonov term keeps the system regular. One LU factorization gives the displacement sensitivities $\boldsymbol{X} = \partial \boldsymbol{q}/\partial \boldsymbol{u}$ and the COM sensitivity $\boldsymbol{S} = \frac{1}{m_{\mathrm{tot}}} \sum_i m_i \boldsymbol{X}_i$. Equivalently,

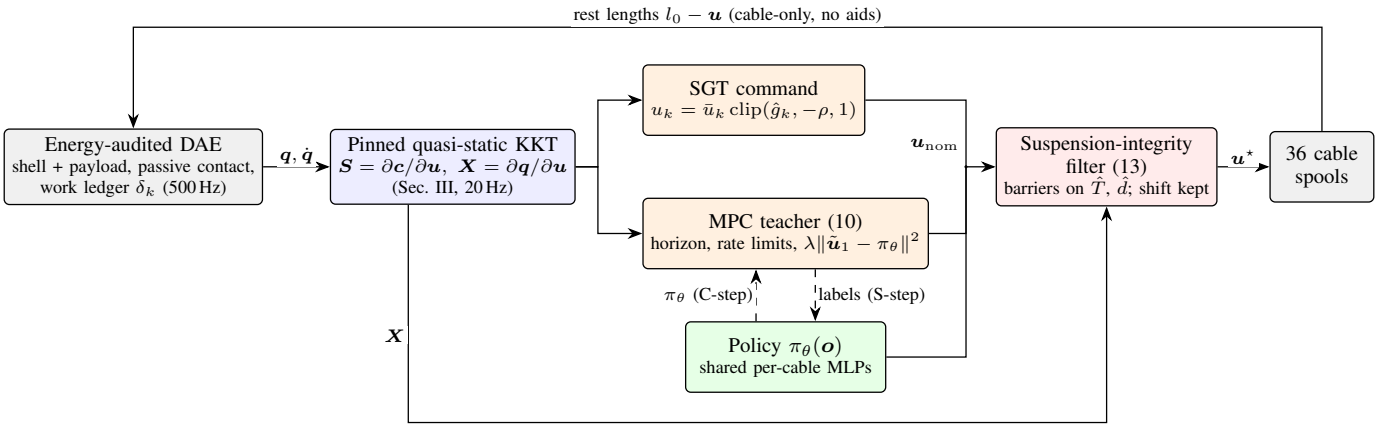


Fig. 2. Architecture. The energy-audited simulator (Sec. II) provides the state; one factorization of the pinned quasi-static KKT system gives the sensitivities used by all three components: SGT (Sec. III), the MPC teacher of MPC + GPS and its learned policy (Sec. IV; dashed: training), and the suspension-integrity filter applied to any nominal command (Sec. V). The rover is actuated only through cable rest lengths.

with $\boldsymbol{Z}$ a basis of $\ker \boldsymbol{A}$ (motions that keep the bars rigid and the support pinned),

$$\boldsymbol{X} = \boldsymbol{Z}\,\boldsymbol{K}_r^{-1}\boldsymbol{Z}^\top \boldsymbol{B}, \qquad \boldsymbol{K}_r = \boldsymbol{Z}^\top \boldsymbol{K}\,\boldsymbol{Z}, \tag{7}$$

where $\boldsymbol{K}_r$ is the reduced stiffness of the admissible motions. Writing $\boldsymbol{K} = \boldsymbol{K}_0 - \sum_k \frac{P_k}{L_k}\boldsymbol{G}_k$ with the cable part $\boldsymbol{K}_0$ and the bar projectors $\boldsymbol{G}_k \succeq 0$ makes the role of the bar compression explicit.

Proposition 2 (Compression increases compliance): If $\boldsymbol{K}_r(\boldsymbol{P}) \succ 0$ and $\boldsymbol{P} \geq \boldsymbol{0}$, then $\boldsymbol{K}_r(\boldsymbol{P})^{-1} \succeq \boldsymbol{K}_r(\boldsymbol{0})^{-1}$; hence for every cable command $\delta\boldsymbol{u}$ the compliance $\delta\boldsymbol{u}^\top \boldsymbol{B}^\top \boldsymbol{Z}\boldsymbol{K}_r^{-1}\boldsymbol{Z}^\top \boldsymbol{B}\,\delta\boldsymbol{u}$, the work of the cable forces on the admissible motion, is at least that of the uncompressed structure.

Proof: $\boldsymbol{K}_r(\boldsymbol{P}) = \boldsymbol{K}_r(\boldsymbol{0}) - \sum_k \frac{P_k}{L_k}\boldsymbol{Z}^\top \boldsymbol{G}_k \boldsymbol{Z} \preceq \boldsymbol{K}_r(\boldsymbol{0})$, and matrix inversion is operator-antitone on positive-definite matrices. ∎

Ignoring the geometric term therefore systematically underestimates which cables can move the COM; the ablation in Section VI confirms the consequence.

B. Edge Selection and Command

For an edge e of $\mathcal{F}$ with outward normal $\boldsymbol{n}_e$ and midpoint $\boldsymbol{p}_e$, the tipping margin is $m_e = -(\boldsymbol{c} - \boldsymbol{p}_e)^\top \boldsymbol{n}_e$ and the gradient of the COM shift across it is $\boldsymbol{g}_e = \boldsymbol{S}_{xy}^\top \boldsymbol{n}_e$. Under box limits $u_k \in [\underline{u}_k, \bar{u}_k]$ the largest first-order shift is the value of the linear program

$$s_e = \max_{\underline{\boldsymbol{u}} \leq \boldsymbol{u} \leq \bar{\boldsymbol{u}}} \boldsymbol{g}_e^\top \boldsymbol{u} = \textstyle\sum_k \max(g_{e,k}\bar{u}_k,\ g_{e,k}\underline{u}_k), \tag{8}$$

which is separable, so its solution is a box vertex. The edge tips to first order once $m_e - \boldsymbol{g}_e^\top(\boldsymbol{u} - \boldsymbol{u}_0) < 0$. Among edges aligned with the travel direction ($\boldsymbol{n}_e^\top \boldsymbol{d} \geq 0.25$), SGT locks for the cycle the edge with the largest reachable margin $s_e - m_e$ and commands

$$u_k = \bar{u}_k \operatorname{clip}\big(g_{e,k}/\|\boldsymbol{g}_e\|_\infty, -\rho, 1\big), \tag{9}$$

normalized separately for shell and payload cables. A state machine (SETTLE→SHIFT→TIP→RELEASE) re-plans at 20 Hz until the support face changes. A support-face primitive contracts a fixed set whose effect is mostly absorbed by elastic deformation; the constrained sensitivity instead tells which cables move the COM, and the compressed bars soften the structure in exactly the bending directions that tipping needs.

IV. MPC + GPS: MPC-Guided Policy Search

A. MPC Teacher

With $\hat{\boldsymbol{g}}$ the gradient of (9) normalized per group and $\tilde{\boldsymbol{u}} = \boldsymbol{u}/\bar{\boldsymbol{u}}$, the teacher solves, over N steps of Δt,

$$\begin{aligned} \min_{\tilde{\boldsymbol{u}}_{1:N}}\ & \textstyle\sum_{k=1}^{N} \big(-\hat{\boldsymbol{g}}^\top \tilde{\boldsymbol{u}}_k + \frac{r}{2}\|\tilde{\boldsymbol{u}}_k\|^2 + \frac{r_\Delta}{2}\|\tilde{\boldsymbol{u}}_k - \tilde{\boldsymbol{u}}_{k-1}\|^2 \big) \\ & + \tfrac{\lambda}{2}\|\tilde{\boldsymbol{u}}_1 - \pi_\theta(\boldsymbol{o})\|^2 \\ \text{s.t. } & \underline{\tilde{u}} \leq \tilde{\boldsymbol{u}}_k \leq 1, \quad |\tilde{\boldsymbol{u}}_k - \tilde{\boldsymbol{u}}_{k-1}| \leq \rho_s, \end{aligned} \tag{10}$$

with $\tilde{\boldsymbol{u}}_0$ the current spool state and ρ_s the spool speed times Δt ($N = 6$, $\Delta t = 50$ ms, $r = 1$, $r_\Delta = 0.2$). The problem is separable per actuator and is solved by projected gradient at 20 Hz.

Proposition 3 (Teacher and SGT): (i) For $N = 1$, $r = 1$, $r_\Delta = \lambda = 0$ and an inactive rate bound, the solution of (10) is the SGT command (9). (ii) In general (10) is separable over actuators and r-strongly convex in each actuator's trajectory with gradient Lipschitz constant $L \leq r + 4r_\Delta + \lambda$; projected gradient with step $1/L$ converges linearly at rate $1 - r/L$.

Proof: (i) Each actuator minimizes $-\hat{g}_k\tilde{u}_k + \frac{1}{2}\tilde{u}_k^2$ over $[\underline{\tilde{u}}_k, 1]$; the unconstrained minimizer is $\hat{g}_k$ and its projection is $\operatorname{clip}(\hat{g}_k, \underline{\tilde{u}}_k, 1)$. (ii) The Hessian of one actuator's cost is $r\boldsymbol{I} + r_\Delta \boldsymbol{D}^\top \boldsymbol{D} + \lambda \boldsymbol{e}_1 \boldsymbol{e}_1^\top$ with the first-difference matrix $\boldsymbol{D}$, $\|\boldsymbol{D}^\top \boldsymbol{D}\| \leq 4$; the constraints are a closed convex set, and the rate is the standard result for strongly convex smooth objectives. ∎

Remark 1. For Gaussian policies with a common covariance Σ, $D_{\mathrm{KL}}\big(\mathcal{N}(\boldsymbol{\mu}_1, \Sigma)\,\|\,\mathcal{N}(\boldsymbol{\mu}_2, \Sigma)\big) = \frac{1}{2}\|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\|^2_{\Sigma^{-1}}$, so the term $\frac{\lambda}{2}\|\tilde{\boldsymbol{u}}_1 - \pi_\theta\|^2$ is the KL-divergence constraint of guided policy search [11] in Lagrangian form with $\Sigma = \boldsymbol{I}$.

B. Policy Observations

The policy π_θ does not use the KKT model. Each outer cable is described by 20 local features in the frame of the travel direction: midpoint relative to the COM, end-node clearances, signed distance to the locked edge and along it, membership of its end nodes in face and edge, the sign-invariant orientation terms $(\boldsymbol{e}^\top \boldsymbol{n}_e)e_z$, $(\boldsymbol{e}^\top \boldsymbol{n}_e)^2$ and e_z^2, current offset and strain, and global context (COM velocity, gait state, margin, time in state); the payload cables have 19 analogous features. One MLP (2×48 units) is shared by all outer cables and one (2×36) by all payload cables. Writing $\boldsymbol{o} = (\boldsymbol{o}_1, \dots, \boldsymbol{o}_{24}, \boldsymbol{o}_1^c, \dots, \boldsymbol{o}_{12}^c)$ for the per-cable features, the policy is

$$\pi_\theta(\boldsymbol{o}) = \big(f_\theta(\boldsymbol{o}_1), \dots, f_\theta(\boldsymbol{o}_{24}),\, f_\theta^c(\boldsymbol{o}_1^c), \dots, f_\theta^c(\boldsymbol{o}_{12}^c)\big). \quad (11)$$

Proposition 4 (Equivariance): Let σ be a symmetry of the six-bar tensegrity that permutes outer and payload cables by $\boldsymbol{P}_\sigma$. Because the features are defined in the frame of the travel direction and the locked edge, they satisfy $\boldsymbol{o}(\sigma\boldsymbol{q}) = \boldsymbol{P}_\sigma \boldsymbol{o}(\boldsymbol{q})$, and (11) gives $\pi_\theta(\boldsymbol{P}_\sigma \boldsymbol{o}) = \boldsymbol{P}_\sigma \pi_\theta(\boldsymbol{o})$.

Proof: Each output depends only on its own feature vector through a shared map, so permuting the inputs permutes the outputs identically. ∎

The six-bar shell has pyritohedral symmetry (order 24), which acts transitively on its 8 closed and on its 12 open triangular faces, so one learned map covers all support faces of the same type; a network on the concatenated state would have to learn every face separately.

C. Training

Starting from $\mathcal{D}_0$ collected with the unregularized teacher, iteration i alternates

$$\text{C-step: roll out (10) with } \lambda_i, \text{ and } \pi_{\theta_i} \text{ itself,} \quad (12a)$$

$$\mathcal{D}_{i+1} = \mathcal{D}_i \cup \{(\boldsymbol{o}, \tilde{\boldsymbol{u}}^\star_{\lambda=0}(\boldsymbol{o}))\}, \quad (12b)$$

$$\text{S-step: } \theta_{i+1} = \arg\min_\theta \textstyle\sum_{\mathcal{D}_{i+1}} \|\pi_\theta(\boldsymbol{o}) - \tilde{\boldsymbol{u}}^\star\|^2, \quad (12c)$$

i.e. every visited state is labelled with the unregularized teacher as in PLATO [14], while the executed actions keep the data close to the learner's own state distribution. We use $\lambda_i \in \{0.5, 1, 2, \dots\}$, four 15 s episodes per iteration (two of them executing the policy) over Levels 1 and 3 with straight and $\pm$8–10 m offset goals, and 8 iterations (114k labelled actuator samples, Adam).

D. Deployment

At run time the policy replaces the per-tick KKT solve while the state machine is kept. Choosing the edge by alignment alone sent the policy to edges it could not tip over, so the deployed controller solves (6) once per tip cycle to pick the edge with the largest reachable margin, as the teacher does, and excludes edges that aborted from the same face.

V. Suspension-Integrity Filter

A. Observation

Gravity and the outer-cable actuation compress the shell while it rolls: under SGT the mean node height drops from 0.467 to 0.394 m (16 %). The node-to-payload distances shrink with it, and the payload cables, pretensioned to 50 N, carry 5.7 N on average; at least one of the 12 is slack in 100 % of the samples (Fig. 1e). The "suspended" payload is loose in the cage for most of the motion.

B. Filter

Given the command $\boldsymbol{u}_{\text{nom}}$ of any tipping controller, the filter returns

$$\begin{aligned} \boldsymbol{u}^\star = \arg\min_{\boldsymbol{u}}\ & \tfrac{1}{2}(\boldsymbol{u}-\boldsymbol{u}_{\text{nom}})^\top \boldsymbol{W}(\boldsymbol{u}-\boldsymbol{u}_{\text{nom}}) \\ \text{s.t. } & \hat{h}_c(\boldsymbol{u}) \ge (1-\gamma)\, h_c, \quad c \in \mathcal{C}, \\ & \hat{\boldsymbol{g}}^\top(\boldsymbol{u}-\boldsymbol{u}_0) \ge \hat{\boldsymbol{g}}^\top(\boldsymbol{u}_{\text{nom}}-\boldsymbol{u}_0), \quad \underline{\boldsymbol{u}} \le \boldsymbol{u} \le \bar{\boldsymbol{u}}, \end{aligned} \quad (13)$$

where $\boldsymbol{u}_0$ is the current command and $\hat{h}_c(\boldsymbol{u}) = h_c + \nabla_{\boldsymbol{u}} h_c^\top(\boldsymbol{u}-\boldsymbol{u}_0)$ is the first-order prediction of a safety function from $\boldsymbol{X}$. For a cable a between nodes i, j with stiffness k, rest length $l_0 - u_a$, length L and direction $\boldsymbol{e}$, the tension $T = k(L - l_0 + u_a)$ has

$$\frac{\partial T}{\partial u_b} = k\Big(\boldsymbol{e}^\top\big(\boldsymbol{X}_{j,b} - \boldsymbol{X}_{i,b}\big) + \delta_{ab}\Big), \quad (14)$$

the first term from the deformation of the shell and the second from the cable's own spool. For the payload at $\boldsymbol{c}$ and rod (i, j) with closest point $\boldsymbol{p}(s) = (1-s)\boldsymbol{q}_i + s\boldsymbol{q}_j$ and distance d,

$$\frac{\partial d}{\partial u_b} = \hat{\boldsymbol{n}}^\top\big(\boldsymbol{X}_{c,b} - (1-s)\boldsymbol{X}_{i,b} - s\boldsymbol{X}_{j,b}\big), \quad \hat{\boldsymbol{n}} = \frac{\boldsymbol{c}-\boldsymbol{p}(s)}{d}, \quad (15)$$

where s is held fixed because $\partial d/\partial s = 0$ at an interior minimizer (envelope theorem). $\mathcal{C}$ contains payload-cable tension $T \ge 10$ N, outer-cable tension $T \le 150$ N and payload-to-rod clearance $d \ge 0.08$ m (core radius 0.05 m), the latter linearized at the closest point of each rod. The barrier condition is the discrete-time form of [17] with $\gamma = 0.3$.

Proposition 5 (Invariance under the prediction): If the QP (13) is feasible at every step and the next value of each safety function equals its prediction, $h_{c,k+1} = \hat{h}_{c,k}(\boldsymbol{u}_k^\star)$, then $h_{c,k} \ge (1-\gamma)^k h_{c,0}$. Hence $h_{c,0} \ge 0$ implies $h_{c,k} \ge 0$ for all k, and an initial violation decays at least geometrically, $\min(0, h_{c,k}) \ge (1-\gamma)^k \min(0, h_{c,0})$.

Proof: Induction on $h_{c,k+1} = \hat{h}_{c,k}(\boldsymbol{u}_k^\star) \ge (1-\gamma)h_{c,k}$ with $0 < 1-\gamma < 1$. ∎

The second constraint preserves the predicted COM shift: without it the filter cancels the payload-cable slackening that SGT uses to tip; with it the tipping effort is re-routed, mainly to the leading payload cables. It is dropped if it conflicts with the barriers. $\boldsymbol{W}$ weights outer cables ten times more than payload cables. Writing the inequality rows of (13) as $\boldsymbol{G}\boldsymbol{u} \ge \boldsymbol{h}$, the QP is strongly convex, so its solution is unique,

and for diagonal $\boldsymbol{W}$ the Lagrangian is minimized over the box in closed form:

$$\boldsymbol{u}(\boldsymbol{\nu}) = \mathrm{clip}\left(\boldsymbol{u}_{\mathrm{nom}} + \boldsymbol{W}^{-1}\boldsymbol{G}^{\top}\boldsymbol{\nu},\ \underline{\boldsymbol{u}},\ \bar{\boldsymbol{u}}\right), \tag{16a}$$

$$\nabla\varphi(\boldsymbol{\nu}) = \boldsymbol{h} - \boldsymbol{G}\boldsymbol{u}(\boldsymbol{\nu}), \qquad L_{\varphi} = \lambda_{\max}\left(\boldsymbol{G}\boldsymbol{W}^{-1}\boldsymbol{G}^{\top}\right), \tag{16b}$$

for the concave dual φ with multipliers $\boldsymbol{\nu} \geq \boldsymbol{0}$. We maximize φ by projected gradient with Nesterov momentum (FISTA [20], $O(1/k^2)$ in the dual), with rows scaled to unit $\boldsymbol{W}^{-1}$-norm.

Remark 2. Proposition 5 holds for the quasi-static prediction $\hat{h}$. The prediction error $h_{c,k+1} - \hat{h}_{c,k}(\boldsymbol{u}_k^{\star})$ is small during quasi-static shifting but not during tumbling and impacts, which are inertial; the closed-loop results of Section VI-E are therefore empirical, and we report where they fail.

VI. Results

A. Experimental Setup

Platform. Shell radius 0.5 m (bar length 0.851 m), 12 nodes of 0.2 kg, six bars of 0.3 kg, payload 1.6 kg (5.8 kg in total), cable stiffness 1200 N/m (payload cables 1600 N/m). Every method uses the same actuator box: stroke 0.15 m (30 % of the 0.49 m rest length) and 0.20 m/s spool speed. Physics runs at 500 Hz, controllers at 20 Hz, on the DAE core with (3), passive contact and zero aids, on terrain Level 1 for 20 s unless stated otherwise.

Baselines. Support-face heuristic: the fixed contraction primitive. *LQR, iLQR, QP-MPC*: the controllers of the original code base, which solve a two-state reduced-order rolling model for a normalized effort that scales the same primitive. *Legacy+assist*†: the original penalty+SHAKE engine with roll couple, traction, grip and corridor aids, reported as an inadmissible reference.

Scenarios and metrics. Lateral impulse 0.5 m/s at 6 s; push 10 N for 2 s; Gaussian, Laplace and outlier (2 %, 10×) sensor noise; cable stiffness $-20\,\%$, friction $-40\,\%$, payload $+30\,\%$; actuation delay 100 ms. We report mean speed $\bar{v}$, cross-track error $e(t) = x_c(t) - x_c(0)$ (RMS and max), success ($\Delta y \geq 1$ m in 20 s), effort $\int \boldsymbol{u}^{\top}\boldsymbol{u}\,dt$, cost of transport $\mathrm{CoT} = W_{\mathrm{act}}/(mg\Delta y)$, controller time and the ledger quantities of Section II; statistics are mean ± std over seeds 1–5 (nominal) or 1–3. The policy is evaluated on 4 held-out conditions (unseen seeds, Levels 1 and 3, straight and ±6 m goals); the filter on 10 paired conditions (three seeds on Level 1, two on Level 3, push, impulse, payload $+30\,\%$, delay, stiffness $-20\,\%$) with payload-cable slack time and tension, payload rattle (RMS position relative to the shell centroid), 99th-percentile payload acceleration, minimal payload–rod clearance and maximal outer-cable tension, measured after the first second.

B. Physical Consistency

Table II compares the cores under the same SGT command sequence. Three observations stand out. *(i) Rigidity.* The DAE core keeps the bar lengths at 2.3×10^{-11} m; the penalty+SHAKE core reaches 1.3×10^{-7} m and, with the interactive aids, 1.9×10^{-4} m. *(ii) Hidden energy exchange.* To keep the bars rigid, the penalty+SHAKE core removes 6.8×10^{3} J by position projection while its cables do a comparable amount of work, i.e. the cables mostly pump energy into the penalty bars that the projection then discards. The DAE core changes only 7.3×10^{-3} J numerically. With the original locomotion aids, 50 % of the energy input comes from non-physical forces, which explains its high speed (0.522 m/s) in Table I. *(iii) Drift.* Before correction, both cores show a per-step energy mismatch that is first-order discretization error: for seed 1, halving Δt from 2 to 1 ms reduces the accumulated $\sum_k |\delta_k|$ from 15.4 to 7.6 J for the DAE core and from 11.0 to 5.5 J for penalty+SHAKE, and the largest single-step mismatch of the DAE core (at the initial landing) from 7.6×10^{-2} to 1.9×10^{-2} J. The energy row removes this drift every step, so ε stays at the solver tolerance (1.0×10^{-9} J). Without the energy row the residual accumulates to 1.5×10^{1} J. We stress that the energy row corrects an integration error of known order; it does not by itself prove that the model is complete—that is the role of δ_k, which would expose an unbooked force as a non-convergent drift. On rugged Level 7 the corrector occasionally reaches its 10-iteration cap during contact transitions, leaving a transient residual of up to 4×10^{-2} J (0.2 % of E) that is removed in the following steps (Table VII).

C. Cable-Only Locomotion with SGT

1) Nominal: with identical actuator limits and all aids disabled, SGT advances 1.81 m in 20 s (0.091 ± 0.026 m/s, 100 % success; Table I, Fig. 3), about 25 times the speed of the support-face heuristic. The LQR, iLQR and QP-MPC schedulers optimize a scalar effort over the same support-face primitive and therefore inherit its limitation: none of them leaves the launch pad (at most 0.10 m), and their cost of transport is an order of magnitude higher. This is a statement about the primitive, which cannot shift the COM of the 5.8 kg system over an edge within a 0.15 m stroke, rather than about the optimizers. SGT plans in 0.68 ms on average (worst 17.4 ms) in single-threaded JavaScript, well inside its 50 ms control period. Its main weakness is heading: the greedy per-cycle edge choice produces a zig-zag track with a cross-track RMSE of 0.253 m (full ground tracks are in the released run files).

2) Robustness: table III summarizes the stress tests (3 seeds each). SGT keeps rolling after a 0.5 m/s lateral impulse (0.094 m/s) and under a 10 N, 2 s push (0.085 m/s), and is insensitive to $-40\,\%$ friction (0.104 m/s) and $+30\,\%$ payload mass (0.083 m/s). Softer cables ($-20\,\%$) slow it to 0.057 m/s because the same stroke produces smaller COM shifts, and a 100 ms actuation delay to 0.050 m/s (67 % success). Measurement noise is the main weakness: Gaussian noise reduces the speed to 0.065 m/s, while heavy-tailed noise is damaging—Laplace noise to 0.026 m/s (0 % success) and 2 % outliers to 0.034 m/s. SGT uses the measured node positions both to detect the support face and to assemble $\boldsymbol{K}$, so a single corrupted node can trigger a wrong face or edge; a robust state estimator in front of the planner is required before hardware use.

TABLE I
NOMINAL CABLE-ONLY LOCOMOTION (LEVEL 1, 20 S, 5 SEEDS, MEAN ± STD).

Method	Speed $\bar{v}\uparrow$ (m/s)	RMSE$\downarrow$ (m)	max $\lvert e\rvert\downarrow$ (m)	Effort$\downarrow$ (m^{2}s)	CoT$\downarrow$ (–)	$t_{\text{solve}}\downarrow$ (ms)	Success$\uparrow$ (%)	Aid$\downarrow$ (%)
SGT (proposed)	**0.091** ±0.026	0.253 ±0.029	0.473 ±0.000	**0.334** ±0.170	**11.4** ±6.6	0.68 ±0.10	**100**	0
Heuristic	0.004	0.083	0.160	2.059	142.4	**0.47** ±0.25	0	0
LQR	0.002	0.129	0.163	2.529	–	0.65 ±0.06	0	0
iLQR	0.005	0.081	0.168	1.757 ±0.000	111.8	0.90 ±0.41	0	0
QP-MPC	0.004	0.084	0.175	1.859 ±0.000	132.0	0.82 ±0.05	0	0
Legacy+assist†	0.522 ±0.023	0.006 ±0.002	0.017 ±0.006	0.636 ±0.060	1.1 ±0.1	0.15 ±0.02	100	50

All methods use the Lagrangian DAE core, passive contact and identical actuator limits. Effort: $\int \boldsymbol{u}^\top \boldsymbol{u}\, dt$. Success: $\Delta y \geq 1$ m in 20 s. Aid: share of the energy input supplied by non-physical aids. Bold: best admissible value; cross-track errors are not marked because they are only comparable between methods that travel a similar distance. †Original engine with roll couple, traction, grip and corridor aids; not admissible, shown for reference.

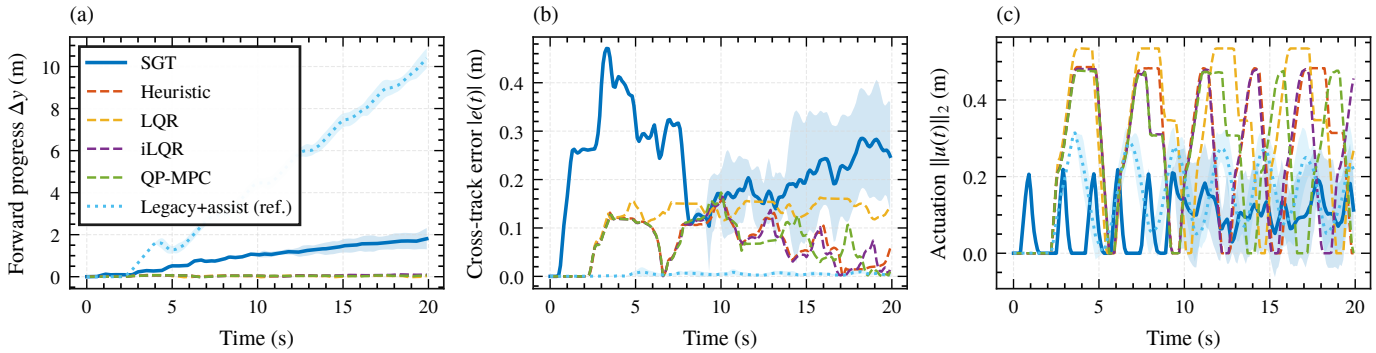


Fig. 3. Nominal cable-only locomotion (5 seeds, mean $\pm 1\sigma$). (a) Forward progress. (b) Cross-track error. (c) Actuation norm. Solid: proposed; dashed: baselines; dotted: inadmissible reference.

TABLE II
PHYSICAL CONSISTENCY OF THE SIMULATOR UNDER THE SGT GAIT (LEVEL 1, 20 S).

Physics core	max $\lvert\varepsilon\rvert$ (J)	max $\lvert\delta_k\rvert\downarrow$ (J)	$\sum_k \lvert\delta_k\rvert\downarrow$ (J)	Bar error$\downarrow$ (m)	$\lvert W_{\text{num}}\rvert\downarrow$ (J)	Speed $\bar{v}$ (m/s)	Aid$\downarrow$ (%)
DAE + Alg. 1, energy row	1.0×10^{-9}	7.6×10^{-2}	1.7×10^{1}	2.3×10^{-11}	7.3×10^{-3}	0.091 ±0.026	0
DAE + Alg. 1, constr. only	1.5×10^{1}	7.1×10^{0}	3.3×10^{1}	5.9×10^{-11}	3.9×10^{1}	0.104 ±0.029	0
Penalty + SHAKE (legacy)	1.1×10^{1}	1.1×10^{-3}	1.1×10^{1}	1.3×10^{-7}	6.8×10^{3}	0.026	0
Legacy + assists†	1.1×10^{1}	1.2×10^{-3}	1.1×10^{1}	1.9×10^{-4}	1.5×10^{4}	0.522 ±0.023	50

$\varepsilon = E - E_0 - \sum W$: energy that no booked work explains *after* correction (held at solver tolerance by the energy row). δ_k: the same quantity for step k *before* correction, i.e. the per-step bound on any unbooked energy source. $|W_{\text{num}}|$: energy changed by constraint projections and by the corrector (booked, but numerical). †Original engine with its locomotion aids and support-face gait.

3) Ablation: table IV isolates the ingredients of SGT (3 seeds). Removing the bars' geometric stiffness is the most harmful change: speed drops from 0.080 to 0.056 m/s, success to 67 % and the cost of transport nearly doubles (23.5 vs. 13.4), confirming that the softening of the compressed bars is what the sensitivity has to capture. The other design choices did not help in our tests: without payload actuation the speed is essentially unchanged (0.076 m/s) at a lower cost of transport (7.7); restricting the command to the 8 most sensitive cables keeps the speed (0.085 m/s) and gives the lowest cost of transport (5.6); and choosing the edge by alignment with the travel direction instead of by reachable margin was not worse (0.102 ± 0.034 m/s). With three seeds these differences are within one standard deviation, so we report the margin-based rule as specified before the experiments and leave the sparse, payload-free variant as the more efficient configuration. Run on the penalty+SHAKE core without aids, the same controller reaches only 0.026 m/s: the projection that absorbs the cable work also absorbs the deformation that SGT relies on. Results on rougher terrain are in the Appendix (Table VII).

D. MPC + GPS

On the held-out conditions the learned policy advances 1.89 m in 20 s, 89 % of SGT (2.11 m) and 83 % of the teacher (2.27 m), with 0.50 aborted cycles per run (Table V), and exceeds SGT in one condition (2.59 vs. 2.35 m). The unregularized teacher is itself slightly faster than SGT, which we attribute to its rate-aware smoothing. The policy is not cheaper per tick in our implementation (0.73 vs. 0.53 ms): for 13 nodes the KKT solve is already inexpensive, so the value of

TABLE III
ROBUSTNESS OF SGT (LEVEL 1, 20 S, 3 SEEDS).

Scenario	SGT $\bar{v}\uparrow$ (m/s)	Heur. $\bar{v}$ (m/s)	SGT RMSE$\downarrow$ (m)	$t_{\mathrm{rec}}\downarrow$ (s)	Success$\uparrow$ (%)
Reference					
Nominal	0.080 $\pm$0.029	0.004	0.258 $\pm$0.040	–	100
Disturbance					
Lateral impulse 0.5 m/s	0.094 $\pm$0.017	0.004	0.266 $\pm$0.011	0.09	100
Push 10 N for 2 s	0.085 $\pm$0.006	$-$0.001	0.308 $\pm$0.035	2.00	100
Sensor noise					
Gaussian	0.065 $\pm$0.016	0.001 $\pm$0.002	0.219 $\pm$0.030	–	67
Laplace	0.026 $\pm$0.016	$-$0.001 $\pm$0.004	0.158 $\pm$0.036	–	0
Outliers (2 %, 10$\times$)	0.034 $\pm$0.023	0.002 $\pm$0.002	0.167 $\pm$0.052	–	33
Plant mismatch					
Cable stiffness $-$20 %	0.057 $\pm$0.002	0.002	0.226 $\pm$0.004	–	100
Friction $-$40 %	0.104 $\pm$0.024	0.002	0.224 $\pm$0.051	–	100
Payload mass +30 %	0.083 $\pm$0.016	0.002	0.249 $\pm$0.022	–	100
Actuation					
Delay 100 ms	0.050 $\pm$0.003	0.004	0.192 $\pm$0.002	–	67

Heur.: support-face heuristic under the same scenario. t_{rec}: time to regain the nominal heading after the disturbance.

TABLE IV
ABLATION OF SGT AND OF THE PHYSICS CORE (LEVEL 1, 20 S, 3 SEEDS).

Variant	Speed $\bar{v}\uparrow$ (m/s)	Tips$\uparrow$	RMSE$\downarrow$ (m)	CoT$\downarrow$ (–)	Success$\uparrow$ (%)
Proposed					
SGT (full)	0.080 $\pm$0.029	8.7 $\pm$3.5	0.258 $\pm$0.040	13.4 $\pm$8.4	**100**
Gait ingredients					
w/o payload actuation	0.076 $\pm$0.005	8.7 $\pm$0.6	0.242 $\pm$0.006	7.7 $\pm$0.0	**100**
w/o feasibility edge choice	0.102 $\pm$0.034	10.3 $\pm$0.6	0.260 $\pm$0.041	8.9 $\pm$3.8	**100**
w/o bar geometric stiffness	0.056 $\pm$0.010	5.7 $\pm$1.2	0.194 $\pm$0.001	23.5 $\pm$4.3	67
sparse: 8 cables	0.085 $\pm$0.002	9.3 $\pm$0.6	0.286 $\pm$0.025	**5.6** $\pm$0.6	**100**
Physics core					
DAE w/o energy row	**0.104** $\pm$0.029	10.3 $\pm$2.1	0.242 $\pm$0.026	9.4 $\pm$5.1	**100**
legacy SHAKE core	0.026	**14.0**	**0.043**	49.1	0

the policy lies in replacing the model evaluation by a learned map, not in speed.

E. Suspension Integrity

With the filter the payload cables are slack 2.7 % of the time under SGT (35.1 % without) and 1.6 % under the policy (33.6 %), in all 10 conditions (Table VI); the mean payload-cable tension rises from 6.2 to 18.0 N (Fig. 1e). For SGT the taut suspension also makes tipping more effective: progress increases from 1.49 to 2.20 m (+48 %, better in 8 of 10 conditions) and aborted cycles fall from 1.1 to 0.0. For the policy the filter restores the suspension at unchanged progress (1.32 vs. 1.29 m). It does not reduce peak loads: in 3 of 20 filtered runs the payload came closer than 0.08 m to a rod (minimum 0.008 m), in 8 runs an outer cable exceeded 150 N, and the 99th-percentile payload acceleration of SGT increased—the inertial events excluded by Remark 2 (prediction error). The filter adds 3.6 ms per control step.

VII. CONCLUSION AND FUTURE PERSPECTIVES

We made tensegrity-rover simulation auditable with a non-minimal Lagrangian DAE, a minimum-norm constraint-and-energy corrector, passive contact with a potential and a work ledger that measures the unexplained energy of every step before it is corrected; it exposes the kilojoule-scale projection exchange and the non-physical aids of a conventional engine. On this basis, sensitivity-gradient tipping rolls a six-bar rover with a heavy payload using cables only, where reduced-order schedulers of a fixed primitive do not. The same sensitivity model serves as the teacher of MPC-guided policy search, whose per-cable policy reaches 89 % of SGT's progress, and as the prediction model of a suspension-integrity filter that keeps the payload cables taut and makes SGT 48 % faster.

Limitations. All results are in simulation with contact and cable parameters not identified on hardware; the ledger bounds the energy contributed by integrator, projections and aids but does not validate the model against reality. SGT is quasi-static, re-plans at 20 Hz (about $0.1\,\mathrm{m/s}$), is sensitive to heavy-tailed measurement noise and zig-zags because its edge choice is greedy per cycle. The policy imitates a teacher on the same model, needs one KKT solve per cycle and is not faster than SGT; the filter guarantees nothing during impacts.

Remark 3. The modelled shell is a regular icosahedron ($d/h = 1/\varphi$). A form-finding analysis with the TsgFEM equilibrium matrix [21] shows that this shape admits no self-stress state, whereas the expanded octahedron ($d/h = 0.5$) admits one; the modelled rest geometry is therefore not a prestressed equilibrium, and part of the observed sag may originate there.

Future perspectives. A second-order barrier on the DAE dynamics for impact-safe payload clearance; repeating the studies on the form-found shape; a robust estimator in front of the planner and a heading-aware edge selection; and hardware validation.

TABLE V
MPC + GPS ON 4 HELD-OUT CONDITIONS (20 S, MEAN $\pm$ STD).

Controller	Progress$\uparrow$ (m)	Tips$\uparrow$	Aborts$\downarrow$	$t_{\mathrm{ctrl}}\downarrow$ (ms)
SGT (9)	2.11 $\pm$0.34	11.0 $\pm$1.4	**0.25** $\pm$0.50	**0.53** $\pm$0.05
MPC teacher (10), $\lambda = 0$	**2.27** $\pm$0.32	10.8 $\pm$1.3	**0.25** $\pm$0.50	0.63 $\pm$0.04
MPC + GPS policy π_θ	1.89 $\pm$0.47	**11.5** $\pm$1.7	0.50 $\pm$0.58	0.73 $\pm$0.05

Held out: unseen seeds, Levels 1 and 3, straight and $\pm$6 m offset goals. Aborts: tip cycles ended by the shift or tip timeout.

REPRODUCIBILITY

`npm run experiments && npm run paper:data && npm run paper:ext` reruns all experiments and regenerates every table, figure and number (https://github.com/kapildev1012/3d).

APPENDIX
TERRAIN GENERALIZATION

Table VII reports SGT and the support-face heuristic on the rougher Levels 4 and 7 (3 seeds per cell). SGT keeps 0.074 m/s on Level 4 and 0.087 m/s on Level 7 with 100 % success; the ledger residual stays below 4×10^{-2} J.

TABLE VI
SUSPENSION-INTEGRITY FILTER ON 10 PAIRED CONDITIONS (20 S, MEAN ± STD OVER CONDITIONS).

	SGT			MPC + GPS policy		
Metric	Filter off	Filter on	Wins	Filter off	Filter on	Wins
Suspension						
Payload-cable slack time (%) ↓	35.1 ±2.9	**2.7** ±0.6	10/10	33.6 ±2.4	**1.6** ±0.6	10/10
Mean payload-cable tension (N) ↑	6.2 ±0.5	**18.0** ±1.0	10/10	6.9 ±0.6	**17.5** ±0.8	10/10
Payload rattle, RMS (mm) ↓	18.6 ±1.0	**15.0** ±2.5	9/10	**22.4** ±1.0	27.5 ±4.6	3/10
Locomotion						
Progress Δy (m) ↑	1.49 ±0.39	**2.20** ±0.64	8/10	**1.32** ±0.64	1.29 ±0.54	4/10
Aborted tip cycles ↓	1.1 ±1.1	**0.0** ±0.0	6/10	**1.5** ±1.4	1.7 ±1.4	3/10
Peak loads (not certified, Remark 2)						
Payload accel., 99th pct (g) ↓	**2.50** ±1.08	3.18 ±1.70	4/10	2.04 ±1.12	**1.71** ±0.66	5/10
Min payload–rod clearance (m) ↑	**0.102** ±0.018	0.090 ±0.035	5/10	**0.105** ±0.014	0.101 ±0.033	4/10
Max outer-cable tension (N) ↓	**137** ±81	179 ±79	3/10	122 ±65	**105** ±52	7/10
Cost						
Controller time (ms) ↓	**0.64** ±0.11	3.60 ±0.21	–	**0.79** ±0.12	3.64 ±0.23	–

Conditions: three seeds on Level 1, two on Level 3, push, impulse, payload $+30\,\%$, 100 ms actuation delay and cable stiffness $-20\,\%$. Wins: paired conditions in which the filter improved the metric. Bold: better mean. Metrics exclude the first second (initial settling).

TABLE VII
TERRAIN GENERALIZATION (3 SEEDS PER CELL).

Method	Speed $\bar{v}$ ↑ (m/s)	RMSE ↓ (m)	Success ↑ (%)	max $\lvert\varepsilon\rvert$ (J)
Level 1 (rolling plain)				
SGT	0.080 ±0.029	0.258 ±0.040	100	1.0×10^{-9}
Heuristic	0.004	0.083	0	9.9×10^{-10}
Level 4 (rocky)				
SGT	0.074 ±0.019	0.275 ±0.044	100	1.0×10^{-9}
Heuristic	0.004	0.083	0	9.9×10^{-10}
Level 7 (rugged relief)				
SGT	0.087 ±0.020	0.229 ±0.008	100	1.5×10^{-2}
Heuristic	0.004	0.083	0	9.9×10^{-10}

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